

Quantum Computation: Independent Study

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May 12, 2026

Abstract

This report presents a structured study of the foundational material in *Quantum Computation and Quantum Information* (Nielsen & Chuang), focusing on Quantum Algorithms and broad understanding. The exposition emphasises conceptual insight and mathematical understanding over mere chapter summary. The fundamentals, the building of Quantum Circuits, usage of Quantum Fourier Transform, and applications on Unstructured Search were studied. Deutsch-Josza Algorithm was implemented using qiskit, for computational exposure. Finally, a paper on Trojan Quantum Walks was read and understood, which helped build a concrete understanding of all concepts learnt so far.

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1 Overview and Motivation: Chapter 1

1.1 The Confluence of Physics and Computer Science

Quantum computation emerged from a productive collision between two apparently unrelated disciplines. Theoretical computer science had, by the 1970s, erected a rigorous edifice around the notion of *computational complexity*—the classification of problems by the resources required to solve them. Meanwhile, physicists studying the thermodynamics of computation (Landauer, Bennett) were asking whether the *physical* laws that govern a computer place any intrinsic limits on what it can compute.

The pivotal insight, due to Feynman (1982) and independently developed by Deutsch (1985), was that the classical Turing machine is an *inefficient simulator* of quantum-mechanical systems. Simulating even a modest n -qubit quantum system on a classical computer requires $\mathcal{O}(2^n)$ memory and time because the state space is exponentially large. A quantum computer, by contrast, manipulates that exponential space *coherently* using only n physical quantum systems.

Key Insight 1.1. The exponential state space of quantum mechanics, ordinarily seen as a complexity burden for simulation, is precisely the resource that quantum computers exploit. A 300-qubit register inhabits a Hilbert space of dimension 2^{300} , far exceeding the number of atoms in the observable universe.

1.2 Information Theory Meets Quantum Mechanics

Classical information theory (Shannon, 1948) quantifies information in bits and studies its transmission, compression, and error correction. Quantum information theory replaces probability distributions with *density matrices* and bits with *qubits*. Several classical results have quantum analogues that are both richer and more constrained:

- **No-cloning theorem:** An unknown quantum state cannot be perfectly copied. This has no classical analogue—a classical bit can be copied freely. It is a direct consequence of the linearity of quantum mechanics: assuming $U |\psi\rangle |0\rangle = |\psi\rangle |\psi\rangle$ for all $|\psi\rangle$ leads immediately to a contradiction.
- **Superdense coding:** Two classical bits can be communicated using a single qubit, *provided* the sender and receiver share a pre-distributed entangled pair. Entanglement thus acts as a communication resource that doubles the capacity of a quantum channel.

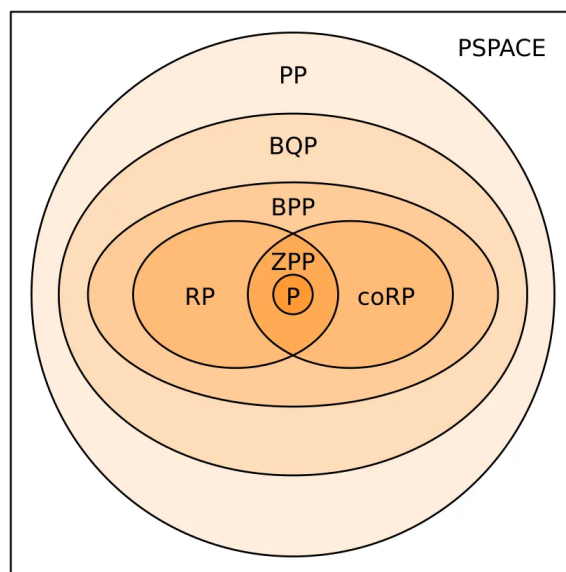


Figure 1: Complexity Classes

- **Quantum teleportation:** An unknown qubit state can be transmitted using two classical bits and one shared EPR pair—no quantum channel needed after the pair is distributed.

1.3 A Roadmap of Quantum Algorithms

Quantum speedups do not arise uniformly—they are tied to specific algebraic or geometric structure in the problem.

Table 1: Selected quantum algorithms and their speedups over the best known classical counterparts.

Algorithm	Problem	Quantum	Classical (best)
Deutsch–Jozsa	Balanced vs. constant function	$\mathcal{O}(1)$ queries	$\mathcal{O}(2^{n-1}) + 1$
Simon’s	Hidden subgroup (mod 2)	$\mathcal{O}(n)$ queries	$\mathcal{O}(2^{n/2})$
Shor’s	Integer factoring	$\mathcal{O}((\log N)^3)$	Sub-exponential
Grover’s	Unstructured search	$\mathcal{O}(\sqrt{N})$	$\mathcal{O}(N)$

The Deutsch–Jozsa and Simon’s algorithms are important not primarily for practical utility but because they were the *first proofs* that quantum computers can solve problems exponentially faster than any deterministic (Deutsch–Jozsa) or randomised (Simon’s) classical algorithm. Shor’s algorithm exploits the Quantum Fourier Transform, which is itself a specialisation of Simon’s hidden subgroup framework. Grover’s algorithm achieves a quadratic—provably optimal—speedup for unstructured search.

1.4 Quantum Teleportation

Quantum teleportation is a protocol that enables the transfer of an arbitrary, unknown quantum state from one location to another using a dual-resource approach: a pre-shared entangled state and a classical communication channel. It is important to note that the physical qubit itself is not moved; rather, the *information* constituting the state is transferred, while the original state is destroyed in the process, satisfying the no-cloning theorem. The Setup The protocol involves three qubits:

1. **Qubit 1 ($|\psi\rangle$):** The source qubit held by Alice, carrying the unknown state $\alpha|0\rangle + \beta|1\rangle$ that she wishes to teleport.
2. **Qubits 2 and 3:** A Bell pair (typically $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$) shared between Alice and Bob. Alice holds Qubit 2, and Bob holds Qubit 3.

The Protocol Steps The process consists of four distinct stages:

1. **Entangling the Source:** Alice performs a CNOT operation on her two qubits (Qubit 1 as control, Qubit 2 as target), followed by a Hadamard gate on Qubit 1. This operation entangles the state to be teleported with Alice's half of the Bell pair.
2. **Measurement:** Alice measures both of her qubits in the computational basis. This measurement yields one of four classical outcomes: 00, 01, 10, or 11. Crucially, this measurement collapses the state of Bob's qubit (Qubit 3) into a state that is related to the original $|\psi\rangle$ by a specific unitary transformation.
3. **Classical Communication:** Alice sends her two bits of classical information to Bob. This step is limited by the speed of light, ensuring that quantum teleportation does not violate causality.
4. **Correction (Recovery):** Bob applies a unitary gate to his qubit based on the bits received from Alice:
 - If 00: Bob applies the Identity (I)—the state is already $|\psi\rangle$.
 - If 01: Bob applies the X gate.
 - If 10: Bob applies the Z gate.
 - If 11: Bob applies ZX (or iY).

1.5 The Deutsch–Jozsa Algorithm

The Deutsch–Jozsa algorithm generalizes the simple Deutsch problem to a function $f : \{0, 1\}^n \rightarrow \{0, 1\}$. We are given the promise that f is either *constant* (returns the same value for all x) or *balanced* (returns 0 for exactly half the possible inputs, and 1 for the other half). Classically, in the worst case, $2^{n-1} + 1$ queries are required to distinguish these cases with certainty. The quantum algorithm requires only a **single query** to the oracle U_f . The procedure is as follows:

1. **Initialise the state:** The starting state is an $(n + 1)$ -qubit register, with the first n qubits in state $|0\rangle$ and the final ancilla qubit in state $|1\rangle$:

$$|\psi_0\rangle = |0\rangle^{\otimes n} |1\rangle \quad (1)$$

2. **Create superposition:** Apply Hadamard gates to all qubits to create a uniform superposition of all possible inputs x and the ancilla in the $|-\rangle$ state:

$$|\psi_1\rangle = \sum_{x \in \{0,1\}^n} \frac{|x\rangle}{\sqrt{2^n}} \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right] \quad (2)$$

3. **Apply the oracle:** The oracle U_f maps $|x\rangle |y\rangle$ to $|x\rangle |y \oplus f(x)\rangle$. Due to the ancilla being in the $|-\rangle$ state, this induces *phase kickback*:

$$|\psi_2\rangle = \sum_x \frac{(-1)^{f(x)} |x\rangle}{\sqrt{2^n}} \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right] \quad (3)$$

Note that the state of the ancilla remains unchanged and can be ignored hereafter.

4. **Perform interference:** Apply a Hadamard transform $H^{\otimes n}$ to the first n qubits. Recall that $H^{\otimes n} |x\rangle = \sum_z \frac{(-1)^{x \cdot z}}{\sqrt{2^n}} |z\rangle$. Thus:

$$|\psi_3\rangle = \sum_z \sum_x \frac{(-1)^{x \cdot z + f(x)}}{2^n} |z\rangle \quad (4)$$

5. **Measurement:** Measure the first n qubits. We consider the probability amplitude of the state $|0\rangle^{\otimes n}$ (where $z = 0$):

$$\text{Amplitude of } |0\rangle^{\otimes n} = \frac{1}{2^n} \sum_x (-1)^{f(x)} \quad (5)$$

Analysis:

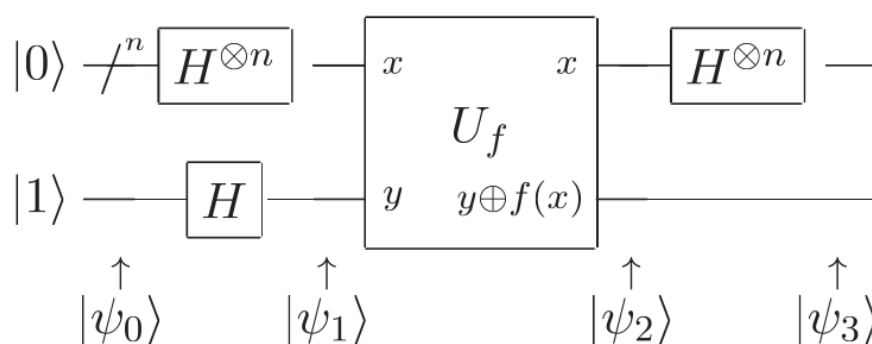


Figure 1.20. Quantum circuit implementing the general Deutsch-Jozsa algorithm. The wire with a '/' through it represents a set of n qubits, similar to the common engineering notation.

Figure 2: Deutsch-Jozsa Circuit

- If f is **constant**, the terms $(-1)^{f(x)}$ are all identical (either all $+1$ or all -1). The sum equals $\pm 2^n$, making the total amplitude ± 1 . We measure $|0\rangle^{\otimes n}$ with probability 1.
- If f is **balanced**, exactly half the terms in the sum are $+1$ and half are -1 . The sum equals 0. The probability of measuring $|0\rangle^{\otimes n}$ is 0, and we must obtain at least one non-zero bit in our measurement.

2 Quantum Mechanics as a Computational Framework: Chapter 2

2.1 Linear Algebraic Structure of Quantum Operations

Chapter 2 of Nielsen & Chuang develops the linear algebraic machinery that underlies every calculation in quantum computation. This subsection collects the results that are most operationally significant, emphasising what each structure *does* in a computation rather than its definition alone.

2.1.1 The Spectral Theorem and Diagonalisation

The most important structural theorem in the chapter is:

Theorem 2.1 (Spectral decomposition). Any **normal** operator M on a finite-dimensional

Hilbert space (i.e. $MM^\dagger = M^\dagger M$) can be written as

$$M = \sum_i \lambda_i |i\rangle \langle i|, \quad (6)$$

where $\{\lambda_i\}$ are the eigenvalues of M and $\{|i\rangle\}$ are an orthonormal set of eigenvectors.

The class of normal operators includes Hermitian operators ($M = M^\dagger$, real eigenvalues), unitary operators ($MM^\dagger = I$, eigenvalues on the unit circle), and positive operators ($\langle \psi | M | \psi \rangle \geq 0$ for all $|\psi\rangle$, non-negative real eigenvalues). All quantum gates are unitary; all observables are Hermitian; all density matrices are positive. The spectral theorem therefore applies to every object encountered in quantum computation.

The spectral decomposition (6) says that every normal operator is, in the right basis, just a list of scalar multiplications. This makes diagonalisation the primary computational tool: if M is hard to work with in the standard basis, find its eigenbasis and the problem becomes trivial. Phase estimation (Section 4.2) does exactly this—it uses quantum interference to find the eigenbasis of a unitary U without ever diagonalising U classically.

2.1.2 Commutators and Their Computational Role

The **commutator** $[A, B] = AB - BA$ and **anti-commutator** $\{A, B\} = AB + BA$ appear throughout Chapter 2. Their computational significance is:

- **Simultaneous diagonalisability.** Two Hermitian operators A and B share a common eigenbasis if and only if $[A, B] = 0$. In practice: two observables can be measured simultaneously without disturbing each other if and only if they commute. The Pauli operators satisfy $[X, Y] = 2iZ$ (and cyclic permutations), so they are *incompatible observables*—measuring X disturbs any prior knowledge of Y or Z .
- **Heisenberg uncertainty.** The generalised uncertainty relation $\sigma_A \sigma_B \geq \frac{1}{2} |\langle [A, B] \rangle|$ quantifies the incompatibility of two measurements in terms of their commutator. For position and momentum, $[x, p] = i\hbar$, giving the standard $\sigma_x \sigma_p \geq \hbar/2$.

2.2 The Four Postulates: A Computational Reading

Rather than recapitulating standard quantum mechanics, this section reinterprets the four postulates as *operational rules for computation*.

Postulate 1 (State space). The state of a closed quantum system is a unit vector $|\psi\rangle$ in a complex Hilbert space $\mathcal{H}$. *Computationally:* a register of n qubits has state space $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$, dimension 2^n .

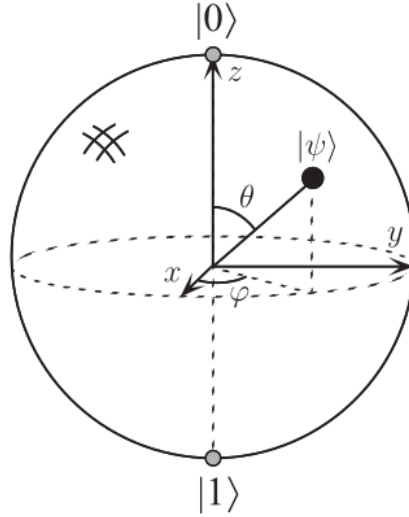


Figure 3: Bloch-sphere representation of a single qubit

Postulate 2 (Evolution). The state evolves via a unitary operator: $|\psi'\rangle = U |\psi\rangle$, with $UU^\dagger = I$. *Computationally*: all quantum gates must be unitary—hence reversible. Classical NAND and AND gates are irreversible and therefore cannot directly serve as quantum gates.

Postulate 3 (Measurement). A measurement is described by operators $\{M_m\}$ satisfying $\sum_m M_m^\dagger M_m = I$. Outcome m occurs with probability $p(m) = \langle\psi| M_m^\dagger M_m |\psi\rangle$, and the post-measurement state is $M_m |\psi\rangle / \sqrt{p(m)}$. *Computationally*: measurement is the *only* irreversible step in a quantum computation. The *deferred measurement principle* states that all measurements can always be moved to the very end of any circuit without changing the distribution of final outcomes.

Postulate 4 (Composite systems). The state space of a composite system is the tensor product of component spaces. *Computationally*: this is what makes quantum computation exponential. Two n -qubit registers are described jointly in a 2^{2n} -dimensional space; their joint state may be *entangled*—not factorisable as $|\psi_A\rangle \otimes |\psi_B\rangle$.

2.3 The Qubit: Geometry and Intuition

A single qubit has state space $\mathbb{C}^2$, and after normalisation and global phase removal, its state space is topologically a 2-sphere—the **Bloch sphere**:

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle, \quad \theta \in [0, \pi], \varphi \in [0, 2\pi). \quad (7)$$

The Bloch sphere makes several facts geometrically transparent:

- The Pauli operators X, Y, Z generate π -rotations about the x, y, z axes respectively.
- The Hadamard gate $H = (X + Z)/\sqrt{2}$ is a π -rotation about the axis bisecting x and z , exchanging $|0\rangle \leftrightarrow |+\rangle$ and $|1\rangle \leftrightarrow |-\rangle$.
- *Orthogonal* states are *antipodal* on the sphere. Since only antipodal states can be distinguished with certainty by a single measurement, two non-orthogonal states are intrinsically indistinguishable with unit probability—a fundamental bound with implications for quantum key distribution.

2.4 Multi-Qubit States and Entanglement

For two qubits, a general state in the computational basis is

$$|\Psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle, \quad \sum_{x \in \{0,1\}^2} |\alpha_x|^2 = 1. \quad (8)$$

A state is **entangled** if and only if it cannot be written as $|\psi_A\rangle \otimes |\psi_B\rangle$. The four **Bell states** are maximally entangled and form an orthonormal basis for the two-qubit space:

$$|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle), \quad (9)$$

$$|\Psi^\pm\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle). \quad (10)$$

They are prepared by applying a Hadamard to the first qubit followed by a CNOT:

$$|\Phi^+\rangle = (\text{CNOT})(H \otimes I)|00\rangle. \quad (11)$$

Remark 2.1. Entanglement is a property of a *partition* of the system, not a property of a single state in isolation. The same four-qubit state may be entangled across one bipartition and separable across another. This relativity of entanglement to partition is essential for understanding multi-qubit algorithms.

2.5 Superdense Coding

Superdense coding is a fundamental quantum communication protocol that demonstrates the remarkable power of entanglement. It allows a sender (traditionally named Alice) to transmit two classical bits of information to a receiver (Bob) by sending only one physical qubit, provided they share a pre-existing entangled state. This effectively doubles the classical capacity of a quantum channel. The Protocol Mechanism The protocol assumes Alice and Bob already share a Bell pair in the state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. Alice holds the

first qubit, and Bob holds the second. To transmit two classical bits $b_1b_2 \in \{00, 01, 10, 11\}$, Alice performs a local operation on her qubit based on the bits she wishes to send:

- **To send 00:** Alice applies the Identity gate (I). The state remains $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.
- **To send 01:** Alice applies the X (NOT) gate. The state becomes $\frac{1}{\sqrt{2}}(|10\rangle + |01\rangle)$.
- **To send 10:** Alice applies the Z (phase-flip) gate. The state becomes $\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$.
- **To send 11:** Alice applies both X and Z (equivalent to iY). The state becomes $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$.

After performing the local operation, Alice sends her single qubit to Bob. Decoding and Measurement Bob now possesses both qubits of the entangled pair. To retrieve the classical bits, he must transform the state back into the computational basis. He performs the inverse of the Bell-state preparation:

1. Apply a CNOT gate (using Alice's qubit as control and his own as target).
2. Apply a Hadamard gate (H) to Alice's qubit.

This measurement in the Bell basis maps the four possible entangled states back to the four classical bitstrings $|00\rangle, |01\rangle, |10\rangle$, and $|11\rangle$ with 100% certainty.

3 Quantum Circuits: Chapter 4

3.1 The Circuit Model and Its Significance

The **quantum circuit model** is the dominant model of quantum computation, playing the same foundational role as Boolean circuits in classical complexity. A circuit is a sequence of quantum gates (unitary operators) applied to an n -qubit register, followed by measurements.

Three design principles are central to circuit-model programming:

1. **Ancilla qubits:** Extra workspace qubits initialised to $|0\rangle$ may be introduced to implement controlled operations or store intermediate results.
2. **Uncomputation:** Any reversible computation $x \mapsto (x, f(x))$ can be run in reverse to restore ancillas to $|0\rangle$, preventing unwanted entanglement with garbage registers.
3. **Deferred measurement:** All measurements can be moved to the end of the circuit without changing output distributions, simplifying correctness analysis.

3.2 Single-Qubit Gates and Their Geometry

Every single-qubit unitary is a rotation of the Bloch sphere. The most important gates are summarised below:

Table 2: Fundamental single-qubit gates.

Gate	Matrix	Bloch action	Primary role
X	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	π about x	Bit flip
Z	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	π about z	Phase flip
H	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	π about $(x+z)/\sqrt{2}$	Creates superposition
S	$\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$	$\pi/2$ about z	Phase gate
T	$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$	$\pi/4$ about z	$\pi/8$ gate
$R_{\hat{n}}(\theta)$	$e^{-i\theta\hat{n}\cdot\vec{\sigma}/2}$	θ about $\hat{n}$	General rotation

An important decomposition theorem underpins single-qubit universality:

Theorem 3.1 (Euler Decomposition). Any $U \in SU(2)$ can be written as

$$U = e^{i\alpha} R_z(\beta) R_y(\gamma) R_z(\delta) \quad (12)$$

for real angles $\alpha, \beta, \gamma, \delta$.

This shows that R_y and R_z rotations—or equivalently the set $\{H, T\}$ —suffice to approximate any single-qubit gate.

3.3 Two-Qubit Gates and Controlled Operations

Two-qubit interactions are necessary to generate entanglement. The canonical two-qubit gate is the **controlled-NOT (CNOT)**:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (13)$$

The CNOT flips the *target* qubit if and only if the *control* qubit is $|1\rangle$.

$$C^n(U) |x_1 x_2 \dots x_n\rangle |\psi\rangle = \begin{cases} |x_1 x_2 \dots x_n\rangle \otimes U |\psi\rangle & \text{if } x_1 = x_2 = \dots = x_n = 1 \\ |x_1 x_2 \dots x_n\rangle \otimes |\psi\rangle & \text{otherwise} \end{cases}$$

3.4 Universal Quantum Gate Sets

Definition 3.1 (Universality). A set of gates $\mathcal{G}$ is **universal** for quantum computation if every n -qubit unitary $U \in U(2^n)$ can be approximated to arbitrary precision (in operator norm) by a circuit composed of gates from $\mathcal{G}$.

Theorem 3.2 (Nielsen & Chuang, Theorem 4.5). The gate set $\{H, T, \text{CNOT}\}$ is universal for quantum computation.

The proof proceeds in two stages:

1. *Single-qubit universality of $\{H, T\}$* : The group generated by H and T is dense in $SU(2)$ because the rotation angles they generate are incommensurable multiples of π . There is no finite subgroup of $SU(2)$ that contains both, so the generated group is infinite and dense.
2. *Multi-qubit extension via CNOT*: Any two-qubit unitary can be decomposed into at most three CNOTs and single-qubit gates; by induction, any n -qubit unitary can be built from CNOTs and single-qubit gates. Combined with step 1, the set $\{H, T, \text{CNOT}\}$ is universal.

Key Insight 3.1. The T gate ($\pi/8$ gate) is the hardest element of $\{H, T, \text{CNOT}\}$ to implement *fault-tolerantly*. On most error-correcting architectures (surface codes, CSS codes), H , S , and CNOT are transversal—they can be applied without spreading errors between qubits. The T gate instead requires *magic state distillation*, a costly subroutine that dominates the resource overhead of fault-tolerant computation. Minimising T -count is therefore the primary optimisation goal in fault-tolerant circuit compilation.

3.5 The Solovay–Kitaev Theorem

The **Solovay–Kitaev theorem** is one of the most practically important results in quantum circuit theory, guaranteeing efficient compilation of arbitrary unitaries into a discrete gate set:

Theorem 3.3 (Solovay–Kitaev). Let $\mathcal{G}$ be a finite universal gate set closed under inverses. Any $U \in SU(2)$ can be approximated to precision ε in operator norm by a sequence of

at most

$$\mathcal{O}(\log^c \frac{1}{\varepsilon}) \quad (14)$$

gates from $\mathcal{G}$, where $c \approx 2$.

This result is remarkable: approximating a continuous rotation to precision ε costs only *polylogarithmic* overhead in $1/\varepsilon$. Without it, compiling floating-point rotation angles to a discrete gate set could be exponentially expensive.

For multi-qubit systems, a dimensionality argument shows that most n -qubit unitaries require $\Omega(4^n/n)$ gates—an amount that grows doubly exponentially. This is why quantum algorithms that achieve their goals with $\text{poly}(n)$ -depth circuits represent non-trivial structural achievements.

3.6 The Quantum Fourier Transform as a Circuit

The **Quantum Fourier Transform (QFT)** is the quantum analogue of the Discrete Fourier Transform (DFT) and the subroutine underlying Shor’s algorithm and quantum phase estimation. For an n -qubit register:

$$\text{QFT } |j\rangle = \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{2\pi i j k / 2^n} |k\rangle, \quad j \in \{0, 1, \dots, 2^n - 1\}. \quad (15)$$

The classical DFT on $N = 2^n$ points requires $\mathcal{O}(N \log N)$ arithmetic operations via the FFT. The QFT circuit uses only $\mathcal{O}(n^2)$ gates—an exponential improvement in *circuit size*.

The circuit is built from Hadamard gates and controlled phase gates $R_k = \text{diag}(1, e^{2\pi i / 2^k})$:

- Apply H to qubit 1.
- Apply controlled- R_2 (control: qubit 2), controlled- R_3 (control: qubit 3), $\dots$, controlled- R_n (control: qubit n) to qubit 1.
- Repeat the same pattern starting from qubit 2 (omitting controls from qubit 1 which has already been processed).
- After all n qubits are processed, apply SWAP gates to reverse the qubit ordering.

Key Insight 3.2. The QFT does *not* give an exponential speedup for every application of the Fourier transform. One cannot read all 2^n Fourier coefficients from a single circuit run—measurement collapses the register to a single computational basis state. The speedup arises only in algorithms specifically designed to extract useful information from

polynomially many measurements. Phase estimation and order-finding are structured precisely this way: the Fourier transform concentrates probability mass near a particular frequency, which can be read off with high probability in a single measurement.

4 Quantum Algorithms: Chapter 5

Chapter 5 of Nielsen & Chuang is where the theoretical machinery of Chapters 2 and 4 crystallises into concrete algorithmic power. The chapter centres on *quantum phase estimation* (QPE), which is the single most important quantum subroutine, and then shows how QPE reduces several classically hard number-theoretic problems—most famously integer factoring—to problems that quantum computers can solve efficiently. The exposition here emphasises *why* each step works, not merely *what* it computes.

4.1 The Phase Kickback Mechanism

Before discussing QPE in full, it is worth isolating the primitive that makes it possible: *phase kickback*.

Suppose U is a unitary with eigenvalue equation $U|u\rangle = e^{2\pi i\varphi}|u\rangle$. Consider the circuit that applies a *controlled- U* with a control qubit in the state $|+\rangle = H|0\rangle$:

$$\begin{aligned} |+\rangle|u\rangle &= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|u\rangle \\ &\xrightarrow{C-U} \frac{1}{\sqrt{2}}(|0\rangle|u\rangle + |1\rangle U|u\rangle) \\ &= \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\varphi}|1\rangle)|u\rangle. \end{aligned} \tag{16}$$

The eigenstate $|u\rangle$ is *unchanged*; the phase $e^{2\pi i\varphi}$ has been “kicked back” onto the control qubit. The control register now encodes φ as a relative phase, which can be extracted by further processing. Phase kickback is the conceptual core of QPE, the Deutsch–Jozsa algorithm, and every algorithm based on the hidden subgroup framework.

Key Insight 4.1. Phase kickback transfers information from the *eigenstructure* of U into the *amplitudes* of the control register. This is the mechanism by which quantum computers can learn about a unitary’s spectrum without ever measuring the eigenstate directly.

4.2 Quantum Phase Estimation

Definition 4.1 (Phase estimation problem). Given a unitary U and an eigenstate $|u\rangle$ such that $U|u\rangle = e^{2\pi i\varphi}|u\rangle$ with $\varphi \in [0, 1)$ unknown, estimate φ to t bits of precision.

The QPE algorithm uses two registers: a *counting register* of t qubits (all initialised to $|0\rangle$) and an *eigenstate register* holding $|u\rangle$.

Step 1 — Create uniform superposition. Apply $H^{\otimes t}$ to the counting register:

$$|0\rangle^{\otimes t} |u\rangle \xrightarrow{H^{\otimes t} \otimes I} \frac{1}{\sqrt{2^t}} \sum_{j=0}^{2^t-1} |j\rangle |u\rangle. \quad (17)$$

Step 2 — Controlled powers of U . Apply controlled- U^{2^k} for $k = 0, 1, \dots, t-1$, where the k -th qubit of the counting register is the control. Using phase kickback, each application multiplies the $|j\rangle$ term by $e^{2\pi i\varphi j}$:

$$\frac{1}{\sqrt{2^t}} \sum_{j=0}^{2^t-1} e^{2\pi i\varphi j} |j\rangle |u\rangle. \quad (18)$$

The counting register now holds the **quantum Fourier transform of the state** $|\tilde{\varphi}\rangle$ where $\tilde{\varphi} = 2^t\varphi$ is the integer encoding of φ .

Step 3 — Inverse QFT. Apply $\text{QFT}^\dagger$ to the counting register:

$$\text{QFT}^\dagger \left(\frac{1}{\sqrt{2^t}} \sum_{j=0}^{2^t-1} e^{2\pi i\varphi j} |j\rangle \right) = |\tilde{\varphi}\rangle, \quad (19)$$

and measuring the counting register yields $\tilde{\varphi}$ (and hence $\varphi = \tilde{\varphi}/2^t$) with high probability.

4.2.1 Precision and Success Probability

When φ is exactly representable in t bits (i.e. $2^t\varphi \in \mathbb{Z}$), the inverse QFT produces the exact answer with probability 1. In general, let b be the best t -bit approximation to φ . Then the probability of measuring b satisfies

$$P(\text{error} > 2^{-t}) \leq \frac{1}{2(n-1)}, \quad (20)$$

where n is the number of additional bits beyond t used for confidence amplification. In practice, using $t + \lceil \log(2 + 1/(2\varepsilon)) \rceil$ counting qubits guarantees success probability at least $1 - \varepsilon$.

Remark 4.1. The three components of QPE—superposition creation, phase encoding via controlled unitaries, and inverse QFT—each play a precise role. The Hadamards create a “probe” that can interfere. The controlled- U^{2^k} ladder *reads* the phase bit-by-bit in a binary expansion. The inverse QFT *writes* those phase bits into a computational basis state. Understanding QPE this way clarifies why the QFT is the right tool: the counting register after Step 2 is literally the QFT of the answer.

4.3 Order Finding

Definition 4.2 (Order finding). Given integers x and N with $\gcd(x, N) = 1$, find the smallest positive integer r such that $x^r \equiv 1 \pmod{N}$.

Order finding is the quantum subroutine at the heart of Shor’s factoring algorithm. It is solved by QPE applied to the *modular multiplication unitary*:

$$U_x |y\rangle = |xy \bmod N\rangle, \quad y \in \{0, 1, \dots, N-1\}. \quad (21)$$

The eigenstates of U_x are the states

$$|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i s k / r} |x^k \bmod N\rangle, \quad s = 0, 1, \dots, r-1, \quad (22)$$

with eigenvalues $e^{2\pi i s / r}$. QPE on $|u_s\rangle$ thus returns s/r .

The crucial trick: one does not need to prepare a specific $|u_s\rangle$. The uniform superposition $\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle = |1\rangle$ is easily prepared. When QPE is applied to $|1\rangle$, each term $|u_s\rangle$ contributes independently, and the measurement returns some s/r chosen uniformly at random. From s/r (as a decimal), the order r is recovered by the *continued fraction algorithm* in $\mathcal{O}((\log N)^3)$ classical time.

Key Insight 4.2. The order-finding reduction reveals the deep structure of the problem: finding the period of the sequence $x^0, x^1, x^2, \dots \pmod{N}$ is equivalent to finding the frequency of a periodic function, and that is exactly what the Fourier transform is designed to do. The quantum speedup arises because the QFT can identify the period of a 2^n -periodic function from a *single evaluation* of its spectrum, whereas a classical algorithm must evaluate the function $\Omega(\sqrt{r})$ times (by a birthday-paradox argument).

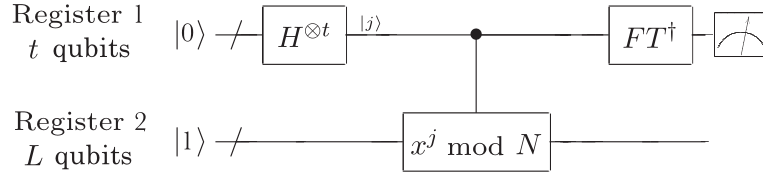


Figure 5.4. Quantum circuit for the order-finding algorithm. The second register is shown as being initialized to the $|1\rangle$ state, but if the method of Exercise 5.14 is used, it can be initialized to $|0\rangle$ instead. This circuit can also be used for factoring, using the reduction given in Section 5.3.2.

Figure 4: Order Finding

4.4 Shor's Factoring Algorithm

Shor's algorithm (1994) factors an n -bit integer N in time $\mathcal{O}(n^3)$, breaking RSA-type cryptography. It reduces factoring to order finding via the following number-theoretic lemma:

Theorem 4.1 (Factoring via order finding). Let $r = \text{ord}_N(x)$ for a random x with $\gcd(x, N) = 1$. If r is even and $x^{r/2} \not\equiv -1 \pmod{N}$, then $\gcd(x^{r/2} \pm 1, N)$ yields a non-trivial factor of N .

The full algorithm is:

- Step 1. Classical pre-processing.** If N is even, return 2. If $N = p^k$ for some prime p , return p (testable classically in $\mathcal{O}(n^3)$). Otherwise, pick a random $x \in \{2, \dots, N-1\}$ and check $\gcd(x, N)$; if > 1 , done.
- Step 2. Quantum order finding.** Use QPE to find $r = \text{ord}_N(x)$. The counting register must have $t = 2n + 1$ qubits to ensure sufficient precision for the continued fraction algorithm to succeed.
- Step 3. Classical post-processing.** If r is odd or $x^{r/2} \equiv -1 \pmod{N}$, go back to Step 1 with a new x . Otherwise compute $p = \gcd(x^{r/2} - 1, N)$ and $q = \gcd(x^{r/2} + 1, N)$.
- Step 4. Verification.** Check that p and q are non-trivial factors of N .

4.4.1 Complexity Analysis

- **Quantum register size:** $t = 2n + 1$ counting qubits plus n qubits for the eigenstate register. Total: $3n + 1$ qubits.
- **Circuit depth:** Each controlled- $U_x^{2^k}$ requires modular exponentiation by 2^k , implementable via repeated squaring in $\mathcal{O}(n^2)$ gates. With $t = 2n + 1$ controlled applications, the total gate count is $\mathcal{O}(n^3)$.

- **Classical post-processing:** The continued fraction algorithm runs in $\mathcal{O}(n^3)$.
- **Repetitions:** The probability that a random x yields an even r with $x^{r/2} \not\equiv -1$ is at least $1 - 1/2^{k-1}$ where k is the number of distinct prime factors of N . For $N = pq$ (RSA case), this probability exceeds $1/2$, so $\mathcal{O}(1)$ repetitions suffice in expectation.

Remark 4.2. The $\mathcal{O}(n^3)$ gate count is for the simplest implementation. Optimised circuits for modular arithmetic (e.g. using the Karatsuba multiplier or the Schönhage–Strassen algorithm) can reduce this to $\mathcal{O}(n^2 \log n \log \log n)$. On a fault-tolerant device, the T -gate count—not the total gate count—is the binding constraint, and recent estimates place the T -count for 2048-bit RSA in the range of 10^{10} – 10^{13} , requiring millions of physical qubits.

4.5 The Discrete Logarithm

QPE and order finding generalise cleanly to the **discrete logarithm problem**:

Definition 4.3 (Discrete logarithm). Given a cyclic group $G = \langle g \rangle$ of order r , and an element $x = g^s$, find the integer $s \in \{0, 1, \dots, r-1\}$.

The quantum algorithm uses a two-register approach. Define two unitaries:

$$U_g |y\rangle = |gy\rangle, \quad U_x |y\rangle = |xy\rangle. \quad (23)$$

Applying QPE *simultaneously* to both registers, with the input $|0\rangle|0\rangle$, produces a joint state from which both the group order r and the logarithm s can be extracted in a single measurement. The running time is $\mathcal{O}(n^3)$, matching Shor’s factoring algorithm.

This is significant: the discrete logarithm underlies Diffie–Hellman and elliptic curve cryptography, and Shor’s algorithm breaks both simultaneously.

5 Quantum Search and Its Limits: Chapter 6

Chapter 6 of Nielsen & Chuang is devoted to **Grover’s quantum search algorithm** and the broader framework of **amplitude amplification**. Where Chapter 5 established exponential speedups for problems with deep algebraic structure, Chapter 6 asks a different question: what can quantum computers do for *unstructured* problems, where no such structure is known? The answer is a provably optimal quadratic speedup—smaller than Shor’s exponential, but universal and rigorously justified.

5.1 The Search Problem and the Oracle Model

Definition 5.1 (Unstructured search). Given a function $f : \{0, 1, \dots, N-1\} \rightarrow \{0, 1\}$ accessible only via a black-box oracle O , find an index ω such that $f(\omega) = 1$. Assume exactly M such marked elements exist, with $M \ll N$.

The oracle is implemented as the unitary map

$$O|x\rangle = (-1)^{f(x)}|x\rangle, \quad (24)$$

which flips the phase of marked states and leaves unmarked states unchanged. This is the *phase oracle* form; it is equivalent (up to a Hadamard on an ancilla) to the *bit oracle* $O|x\rangle|q\rangle = |x\rangle|q \oplus f(x)\rangle$.

Remark 5.1. The oracle model is crucial for proving lower bounds. By treating f as a black box, one counts how many times the oracle must be queried, which captures the inherent difficulty of the problem independent of any implementation. Classical algorithms require $\Omega(N)$ queries in the worst case; Grover's algorithm requires $\mathcal{O}(\sqrt{N})$ queries.

5.2 Grover's Algorithm: Mechanics

Grover's algorithm initialises the n -qubit register (with $N = 2^n$) in the uniform superposition

$$|s\rangle = H^{\otimes n}|0\rangle^{\otimes n} = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle, \quad (25)$$

and then applies the **Grover iterate** G repeatedly:

$$G = D \cdot O, \quad (26)$$

where:

- O is the phase oracle (24), which marks the solution(s) by phase flip.
- $D = 2|s\rangle\langle s| - I$ is the **diffusion operator** (Grover diffusion), which performs *inversion about the mean*.

After k applications of G , the state is measured in the computational basis. The algorithm terminates after $R = \lfloor \pi/(4\theta) \rfloor$ iterations, where $\sin \theta = \sqrt{M/N}$.

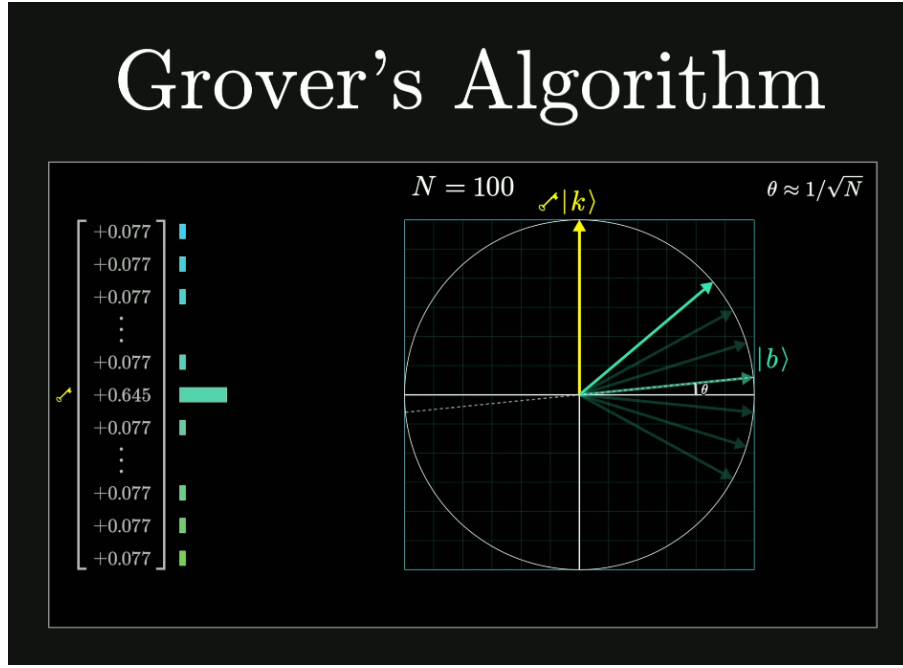


Figure 5: Grover as in Youtube Video 3blue1brown

5.2.1 The Diffusion Operator in Circuit Form

The diffusion operator $D = 2|s\rangle\langle s| - I$ can be implemented as

$$D = H^{\otimes n} \cdot (2|0\rangle\langle 0| - I) \cdot H^{\otimes n}, \quad (27)$$

where $(2|0\rangle\langle 0| - I)$ flips the phase of every basis state *except* $|0\rangle$. This is implemented by a multi-controlled Z gate:

$$(2|0\rangle\langle 0| - I)|x\rangle = \begin{cases} |x\rangle & x = 0, \\ -|x\rangle & x \neq 0. \end{cases} \quad (28)$$

The complete circuit for one Grover iterate is therefore: (i) phase oracle O ; (ii) $H^{\otimes n}$; (iii) controlled phase flip about $|0\rangle$; (iv) $H^{\otimes n}$.

5.3 Success Probability and the Danger of Over-Rotation

From equation (??), after k iterations the probability of measuring a marked state is

$$P(\text{success at step } k) = \sin^2((2k+1)\theta). \quad (29)$$

For $M = 1$ (single marked item), $\theta \approx 1/\sqrt{N}$, and the optimal number of iterations is

$R = \lfloor \pi\sqrt{N}/4 \rfloor$, giving success probability

$$P_{\max} = \sin^2\left((2R+1)\frac{1}{\sqrt{N}}\right) \geq 1 - \frac{1}{N}, \quad (30)$$

which approaches 1 as $N \rightarrow \infty$.

Effect of unknown M . If M is unknown, the optimal stopping point is also unknown. Two strategies are available:

1. **Quantum counting (from Chapter 5):** First run QPE on the Grover iterate to estimate M , then run Grover with the correct R . Total cost: $\mathcal{O}(\sqrt{N})$ queries.
2. **Random stopping:** Run Grover for a uniformly random number of iterations between 1 and some upper bound T . A classical argument shows this finds a solution with constant probability in $\mathcal{O}(\sqrt{N})$ queries even without knowing M .

One of the most important results in Chapter 6 is that Grover’s algorithm is *optimal*: no quantum algorithm can solve unstructured search in fewer than $\Omega(\sqrt{N})$ queries. The proof uses the **hybrid argument** due to Bennett, Bernstein, Brassard, and Vazirani (1997).

6 Implementation Note: Deutsch–Jozsa in Qiskit

This section documents a software implementation of the Deutsch–Jozsa algorithm using IBM’s **Qiskit** framework (v1.x) and the **qiskit-aer** statevector simulator. The implementation was carried out as a practical companion to the theoretical study, with the goal of building familiarity with the quantum circuit model as it is realised in software.

6.1 Software Environment

Qiskit is an open-source Python SDK developed by IBM for programming quantum computers at the level of pulses, circuits, and algorithms. A circuit is constructed as a **QuantumCircuit** object, gates are appended symbolically, and execution is dispatched to either a local simulator (via **qiskit-aer**) or a real IBM quantum device (via IBM Quantum Platform). For this study, all experiments were run on the **AerSimulator** backend, which performs exact statevector simulation.

The choice of Deutsch–Jozsa as the first implementation is deliberate. It is the simplest algorithm that requires only: (i) single-qubit Hadamard gates, (ii) a single oracle query, and (iii) a final measurement. It therefore exercises the minimum viable quantum circuit

without the additional complexity of controlled powers of unitaries (Shor) or iterative amplitude amplification (Grover).

6.2 Circuit Structure

The implementation uses n input qubits and one ancilla qubit (qubit index n), for a total register of $n + 1$ qubits. The circuit proceeds in five stages, each separated by a **barrier** for visual clarity:

1. **Ancilla initialisation.** An **X** gate followed by a **H** gate prepares the ancilla in $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$. This is the standard setup for phase kickback: when the oracle applies a **CNOT** with the ancilla as target, the phase $(-1)^{f(x)}$ is kicked back onto the control qubit rather than flipping the ancilla's state.
2. **Uniform superposition.** **H** gates on all n input qubits produce $|s\rangle = H^{\otimes n} |0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle$.
3. **Oracle query.** The oracle sub-circuit U_f is composed into the main circuit via `QuantumCircuit.compose()`. This cleanly separates the oracle definition from the algorithm logic, reflecting the black-box model precisely.
4. **Second Hadamard layer.** **H** gates are applied again to all n input qubits. This step performs the quantum interference that distinguishes constant from balanced functions.
5. **Measurement.** The n input qubits are measured into n classical bits. The ancilla is never measured; its role ends with the oracle query.

6.3 Oracle Implementations

Three oracle types were implemented, each as an independent `QuantumCircuit` sub-circuit:

Constant oracle ($f(x) = c$). For $c = 0$, the circuit is the identity—no gates are applied. For $c = 1$, a single **X** gate is applied to the ancilla. In both cases the input qubits are untouched, which is consistent with the oracle definition: a constant function carries no information about the input. After phase kickback, the **X** gate for $c = 1$ contributes only a global phase to the ancilla state, leaving the computational state of the input register unchanged.

Balanced parity oracle ($f(x) = x_0 \oplus x_1 \oplus \dots \oplus x_{n-1}$). A CNOT ladder is applied from each input qubit (as control) onto the ancilla (as target). The parity function is balanced because exactly 2^{n-1} of all n -bit strings have even parity and 2^{n-1} have odd parity. The CNOT ladder realises $|x\rangle|q\rangle \mapsto |x\rangle|q \oplus f(x)\rangle$, which in the presence of an ancilla in $|-\rangle$ produces the phase oracle form $(-1)^{f(x)}|x\rangle$ via kickback.

Masked balanced oracle. A more general balanced function is constructed by wrapping the parity CNOT ladder with **X** gates on a selected subset of input qubits (determined by a bitmask), and uncomputing those **X** gates afterwards. This shifts the set of inputs that map to 1 while preserving the balanced property. It was implemented to verify that the algorithm correctly returns “balanced” regardless of *which* balanced function is presented—a direct demonstration that Deutsch–Jozsa distinguishes the two classes but provides no information about the specific function within each class.

6.4 Results and Observations

Four experiments were run, each with $n = 4$ input qubits and 1024 shots. The results are summarised in Table 3.

Table 3: Experimental results for $n = 4$ input qubits on the `AerSimulator` (1024 shots each).

Oracle	Measurement	Shots on outcome	Verdict
Constant, $f(x) = 0$	$ 0000\rangle$	1024 / 1024	Constant
Constant, $f(x) = 1$	$ 0000\rangle$	1024 / 1024	Constant
Balanced (parity)	$ 1111\rangle$	1024 / 1024	Balanced
Balanced (masked)	$ 1111\rangle$	1024 / 1024	Balanced

Every experiment produced a single outcome with frequency 1024/1024, confirming that the algorithm is **deterministic**: unlike Grover’s algorithm, which has a sinusoidal success probability requiring careful iteration counting, Deutsch–Jozsa produces the correct answer with certainty in a single shot. This is a direct consequence of the exact cancellation of amplitudes by quantum interference.

Three observations from the implementation are worth recording:

1. **Both constant oracles produce the same measurement outcome $|00\dots 0\rangle$.** The distinction between $f(x) = 0$ and $f(x) = 1$ is not observable at the output: both are declared “constant.” The algorithm makes no finer distinction, which is exactly what the problem specification requires.

2. **The masked balanced oracle produces the same outcome as the plain parity oracle** ($|1111\rangle$), despite computing a different function. This confirms the theoretical prediction: the output encodes only the class membership (balanced), not the identity of the function. In the query complexity model, the algorithm has learned the maximum possible from one query.
3. **The algorithm scales cleanly.** Changing the parameter n in the implementation automatically adjusts all gate layers and the oracle. For $n = 10$, the circuit contains 23 gates (n Hadamards + n CNOTs in the oracle + n Hadamards + 2 ancilla gates), whereas a classical deterministic algorithm would require up to $2^9 + 1 = 513$ queries. The exponential separation is visible even at these small scales.

6.5 Connecting Implementation to Theory

Several aspects of the implementation illuminate theoretical points that can be abstract when read from the text:

Phase kickback is invisible in the circuit diagram but essential to correctness.

The CNOT gates in the oracle appear to act on the ancilla qubit, yet the ancilla measurement is never taken and its state is irrelevant at the end. What the CNOTs actually accomplish—in the presence of the $|-\rangle$ ancilla—is to *write information into the phases of the input register*. This phase information is then read out by the second Hadamard layer. The circuit diagram does not make this explicit; understanding it requires tracking the state algebraically, as in Section 4.1.

The second Hadamard layer is the decoder. Its role is precisely that of the inverse QFT in the special case of $G = \mathbb{Z}_2^n$: it converts phase information into amplitude information that measurement can access. This makes Deutsch–Jozsa a minimal instance of the general pattern introduced in Chapters 5 and 6: *encode the answer in a phase, then apply a Fourier transform to convert it to a measurable amplitude*.

The oracle is a sub-circuit, not a function call. In Qiskit, the oracle is a `QuantumCircuit` object composed into the main circuit. This mirrors the theoretical model precisely: the oracle U_f is a unitary black box that the algorithm can query but not inspect. Separating the oracle construction from the algorithm logic enforces this boundary in code and makes it straightforward to swap in any oracle without modifying the algorithm.

6.6 Limitations and Extensions

The implementation uses an exact statevector simulator, which maintains the full 2^{n+1} -dimensional complex state vector in classical memory. This is feasible for small n but becomes intractable for $n \gtrsim 30$ on a standard workstation. On a real quantum device, the same circuit would run regardless of n (up to the device qubit count), but would be subject to gate errors, decoherence, and readout noise—effects absent from the simulation.

7 Paper Reading: Trojan Quantum Walks

7.1 The One-Dimensional Discrete Quantum Walk: Model Recap

Ghizoni & Amorim study discrete-time quantum walks on a one-dimensional integer lattice $\mathbb{Z}$. The Hilbert space is

$$\mathcal{H} = \mathcal{H}_C \otimes \mathcal{H}_P, \quad (31)$$

where $\mathcal{H}_C = \mathbb{C}^2$ is the *coin (spin) space*, spanned by $\{|\uparrow\rangle, |\downarrow\rangle\}$, and $\mathcal{H}_P = \ell^2(\mathbb{Z})$ is the *position space*, spanned by $\{|j\rangle\}_{j \in \mathbb{Z}}$.

A general initial state is

$$|\Psi(0)\rangle = \sum_{j=-\infty}^{+\infty} [a(j, 0) |\uparrow\rangle + b(j, 0) |\downarrow\rangle] \otimes |j\rangle, \quad (32)$$

normalised as $\sum_j (|a(j, 0)|^2 + |b(j, 0)|^2) = 1$. The paper considers two families of initial position distributions:

- **Local (delta) state:** $f(j) = \delta_{j,0}$, i.e., the walker starts at the origin. This is the standard initial condition in most of the quantum walk literature.
- **Gaussian state:** $f(j) \propto e^{-j^2/(4\sigma_0^2)}$, with initial spread σ_0 . The walker starts delocalised over many lattice sites, with σ_0 controlling the degree of delocalisation.

One step of time evolution is governed by the operator

$$U(j-r) = S [C(j-r) \otimes |j\rangle\langle j|], \quad (33)$$

where $S = |\uparrow\rangle\langle\uparrow| \otimes |j+1\rangle\langle j| + |\downarrow\rangle\langle\downarrow| \otimes |j-1\rangle\langle j|$ is the **conditional displacement operator**—it shifts $|\uparrow\rangle$ right and $|\downarrow\rangle$ left—and $C(j-r)$ is a **position-dependent quantum coin**:

$$C(j-r) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} + \frac{\delta_{j,r}}{\sqrt{2}} \begin{pmatrix} -1 & \sqrt{2}-1 \\ \sqrt{2}-1 & 1 \end{pmatrix}. \quad (34)$$

For $j \neq r$, (34) reduces to the standard Hadamard coin. At the single special site $j = r$, it becomes the NOT gate σ_x , which swaps $|\uparrow\rangle \leftrightarrow |\downarrow\rangle$.

Key Insight 7.1. The position-dependent coin is a remarkably minimal modification: one site out of infinitely many is changed from a Hadamard to a NOT gate. Yet this local defect has a global effect—it acts as a *quantum mirror*, reflecting the chirality (spin

direction) of the walker at that site and sending probability amplitude propagating in the opposite direction. This is the only non-standard ingredient in the entire paper.

7.2 Standard Quantum Walk Behaviour

To appreciate the Trojan state, it is necessary to first understand what standard (Hadamard coin, no defect) quantum walks do.

The key feature distinguishing a quantum walk from a classical random walk is its **ballistic spreading**: the standard deviation of position grows as $\sigma(t) \propto t$, compared to $\sigma(t) \propto \sqrt{t}$ for a classical random walk. This quadratic improvement in mixing rate is the same speedup seen in Grover's algorithm and is due to quantum interference concentrating probability amplitude near the two extremes of the distribution rather than the centre.

A Hadamard walk from a local initial state produces an *asymmetric* probability distribution (the asymmetry depending on the initial qubit), while a walk from a Gaussian state with large σ_0 concentrates nearly all probability into two sharp, well-separated peaks moving apart at speed $\approx \sqrt{2}$ per time step. As $\sigma_0 \rightarrow \infty$, the distribution becomes perfectly bimodal with no amplitude between the peaks.

The **entanglement between spin and position** is quantified by the von Neumann entropy of the reduced coin state:

$$S_E(t) = -\text{tr}[\rho_C(t) \log_2 \rho_C(t)], \quad \rho_C(t) = \text{tr}_P[|\Psi(t)\rangle \langle \Psi(t)|]. \quad (35)$$

For the standard Hadamard walk, $S_E(t)$ approaches a long-time limit of approximately 0.87 (local state), 0.77 ($\sigma_0 = 1$), and 0.69 ($\sigma_0 = 10$), averaged over all initial qubits. The entanglement between spin and position grows because the coin and displacement operators are *entangling*: the final position of the walker depends on the spin, creating spin-position correlations that the von Neumann entropy quantifies.

7.3 The NOT Gate as a Quantum Mirror

When a NOT gate is placed at position $j = r$, the walker arriving at that site has its spin flipped: $|\uparrow\rangle \rightarrow |\downarrow\rangle$ and vice versa. Since the displacement operator moves $|\uparrow\rangle$ right and $|\downarrow\rangle$ left, a spin flip at $j = r$ reverses the propagation direction of any amplitude that reaches the defect site.

The paper places the NOT gate at $j = r = -101$, so walkers propagating leftward are reflected back rightward. Crucially, this is a *quantum* reflection, not a classical boundary:

the amplitude is not simply turned around, but interferes with the rightward-propagating component of the wavefunction.

7.4 The Trojan Wave Packet

The central result of the paper is the following: when the initial state is a *Gaussian with sufficiently large σ_0* , the reflection of one peak by the NOT gate creates a **Trojan wave packet**—a non-spreading, non-stationary double-peak quantum state.

The mechanism has three stages:

1. **Bimodal preparation.** For large σ_0 , the Hadamard walk quickly develops into two sharp peaks propagating at $\pm\sqrt{2}$ sites per step with negligible amplitude between them. Almost all probability is carried by the two peaks.
2. **Reflection.** The leftward-moving peak hits the NOT gate at $j = -101$ and is reflected. After reflection, it propagates rightward at $+\sqrt{2}$ sites per step, now co-propagating with the original rightward peak. The two peaks are separated by $2 \times 101 = 202$ sites.
3. **Interference and Trojan formation.** Since both peaks now move in the same direction at the same speed, their relative position is fixed. The quantum interference pattern between them cancels the spreading of the distribution: amplitude that would ordinarily leak *between* the peaks interferes destructively, confining the probability to the two isolated peaks. The result is a state that propagates as a rigid double-peak structure—non-spreading and non-stationary—the defining characteristics of a Trojan wave packet.

Key Insight 7.2. The Trojan effect is fundamentally a consequence of the *initial delocalisation*. A local initial state ($\sigma_0 = 0$) does not produce a Trojan packet after reflection—the peaks continue to spread because they have appreciable amplitude between them from the start, and the reflection does not eliminate that inner amplitude. The Gaussian walk with large σ_0 is special because the bimodal structure is already nearly perfect before the reflection event, so the two post-reflection peaks are individually sharp and their mutual interference is constructive only at the peak positions. This is an instance of the general principle that quantum interference is more effective when amplitudes are already spatially organised.

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